11

Chapter 11 01068007 Control Systems

Frequency Response Design

Loop shaping, compensators and fundamental limits

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Lecture outline

Part 1 Frequency-domain specifications

PM, GM, DM, M_S · PM $\rightarrow \zeta \rightarrow M_p$ · three frequency bands · five-step procedure

Part 2 Six compensators

P · PI · Lag · Lead · Lag–lead · PID · analytical design · Nichols

Part 3 Fundamental limitations

Bode gain–phase · waterbed · RHP zero & dead time · IMC · Smith predictor

Part 4 Choosing and checking a design

Design Atlas of 25 plants · checklist · MATLAB

Learning objectives

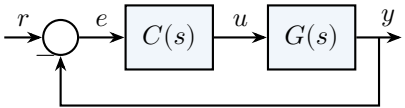
After this chapter you should be able to

1. Translate M_p, t_s, e_{ss} and robustness requirements into bounds on $|L(j\omega)|$, PM, GM, ω_{gc} and M_S
2. Design P, PI, lag, lead, lag–lead and PID controllers with a standard procedure and verify them numerically
3. Obtain closed-form parameters from ω_c and PM, and decide whether PI or PD is feasible
4. Use the Bode gain–phase relation and the sensitivity integral to explain why some specifications are impossible
5. Compute the speed ceilings imposed by RHP zeros and dead time
6. Tune PI for FOPDT processes by IMC, Ziegler–Nichols and frequency-domain methods, and evaluate the Smith predictor
7. Use a checklist to catch loops whose GM/PM look good but are not robust

Part 1

Frequency-Domain Specifications

From analysis to design: loop shaping



$$L = CG, \quad S = \frac{1}{1+L}, \quad T = \frac{L}{1+L}, \quad S + T = 1$$

- Chapter 10: *read* $L(j\omega)$ and decide stability
- Chapter 11: *choose* $C(s)$ so that $L(j\omega)$ has the required margins, speed and accuracy
- Advantages: works with measured frequency data · dead time $e^{-\theta s}$ is handled exactly ($-\omega\theta$) · margins are robustness indices

Stability margins: PM, GM, DM

$$|L(j\omega_{\text{gc}})| = 1, \qquad \text{PM} = 180^\circ + \angle L(j\omega_{\text{gc}})$$
$$\angle L(j\omega_{pc}) = -180^\circ, \qquad \text{GM} = 1/|L(j\omega_{pc})|$$
$$\text{DM} = \text{PM}\,[\text{rad}]/\omega_{\text{gc}}$$

Watch out

- $|L|$ crosses 0 dB several times $\rightarrow$ check PM at *every* crossing
- Plant with RHP poles $\rightarrow$ GM may be a *lower* bound
- DM tells how much extra delay the loop tolerates

Example: PM $= 40^\circ$ at $\omega_{\text{gc}} = 2$ rad/s $\rightarrow$ DM $= 0.349$ s; at $\omega_{\text{gc}} = 20$ rad/s $\rightarrow$ DM is only 0.035 s

Peak sensitivity M_S — the real robustness index

$$M_S = \max_{\omega} |S(j\omega)| = \frac{1}{\min_{\omega} |1 + L(j\omega)|}$$

$1/M_S$ is the shortest distance from the Nyquist curve to -1 in *any* direction — so it covers GM and PM at once

$$\text{GM} \geq \frac{M_S}{M_S - 1}, \qquad \text{PM} \geq 2 \sin^{-1} \frac{1}{2M_S}$$

Targets used in this chapter

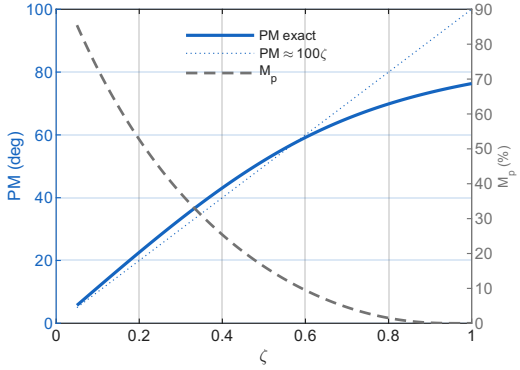
$M_S = 1.4$ conservative · $M_S = 1.6$ balanced · $M_S = 2.0$ aggressive

M_S guarantees minimum GM and PM

M_S	GM $\geq$ (dB)	PM $\geq$
1.2	15.56	49.3°
1.4	10.88	41.9°
1.6	8.52	36.4°
2.0	6.02	29.0°
2.5	4.44	23.1°

The converse is false: good GM *and* PM do not guarantee a small M_S (see the counterexample in Part 4)

From PM to the time response



$$\text{PM} = \tan^{-1} \frac{2\zeta}{\sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}}$$

Table $M_p \to \zeta \to$ PM

M_p	ζ	PM	100ζ	$t_s\omega_{gc}$
5%	0.690	64.6	69.0	3.79
10%	0.591	58.6	59.1	4.89
16.3%	0.500	51.8	50.0	6.29
20%	0.456	48.2	45.6	7.17
30%	0.358	39.1	35.8	9.85

Rule of thumb $\zeta \approx \text{PM}/100$

Exact only at $\zeta = 0.574$ · below that it is optimistic

Steady-state accuracy: set the gain first

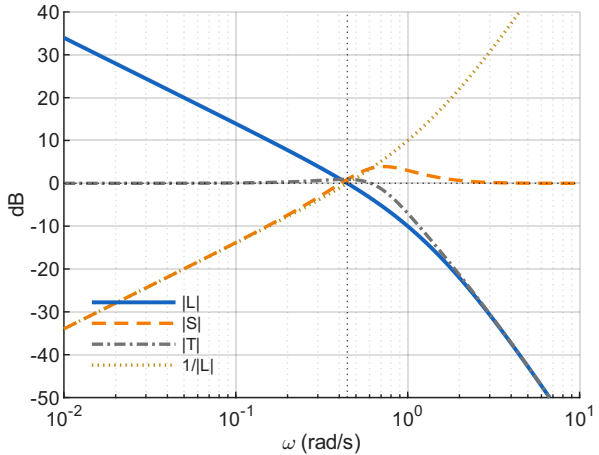
$$K_p = \lim_{s\to 0} L, \quad K_v = \lim_{s\to 0} sL, \quad K_a = \lim_{s\to 0} s^2 L, \qquad e_{ss,\text{step}} = \frac{1}{1+K_p}, \quad e_{ss,\text{ramp}} = \frac{1}{K_v}$$

First step of every design

$$K_c = \frac{K_{\text{req}}}{\lim_{s\to 0} s^N G(s)}$$

- Low-frequency level of $|L|$ is $20\log K_p$ (Type 0) · Type 1: the -20 dB/dec asymptote crosses 0 dB at $\omega = K_v$
- Example: $G = 1/[s(s+1)]$ with $K_v = 10$ required $\rightarrow K_c = 10$

The three frequency bands of a feedback loop



$L = 1/[s(s+1)(s+2)]$, $\omega_{gc} = 0.446$, $M_S = 1.568$

Three bands: how to read the plot

$$y = T\,r + S\,d - T\,n$$

- $|L| \gg 1$: $|S| \approx 1/|L| \to$ good tracking and disturbance rejection
- $|L| \ll 1$: $|T| \approx |L| \to$ noise attenuation, robustness to high-frequency dynamics
- Near crossover: neither approximation holds (0.93 dB error) — this is where stability is decided

Summary

Low-frequency specs = *lower* bounds on $|L|$ · high-frequency specs = *upper* bounds on $|L|$

The five-step design procedure

1. **Specify** PM, GM, ω_{gc}, $e_{ss} \rightarrow$ set K_c
2. **Choose** ω_{gc} (respect the delay/RHP ceilings)
3. **Evaluate** $|G(j\omega_{gc})|$, $\angle G(j\omega_{gc}) \rightarrow \phi_{req} = -180^\circ + \text{PM} - \angle G(j\omega_{gc})$
4. **Shape** C so that $\angle C = \phi_{req}$ and $|CG| = 1$
5. **Verify** PM, GM, M_S, closed-loop poles, step response, $|u|_{max}$

Read the sign of ϕ_{req}

$\phi_{req} \leq 0 \rightarrow$ P, PI or lag is enough $\cdot$ $\phi_{req} > 0 \rightarrow$ lead, PD or PID is needed

Number to remember

A PI/lag zero one decade below ω_{gc} costs only $\tan^{-1} 0.1 = 5.71^\circ$ of phase

Part 2

Six Compensators

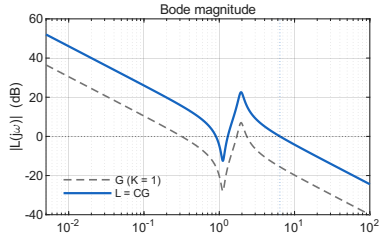
P control: shift the magnitude, leave the phase

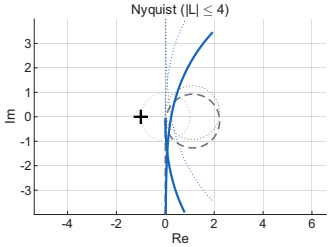
- To set the speed: $K = 1/|G(j\omega_{gc})|$
- To set the PM: find ω where $\angle G = -180^\circ + \text{PM}$, then $K = 1/|G(j\omega)|$
- Impossible for Type-2 plants and undamped resonances

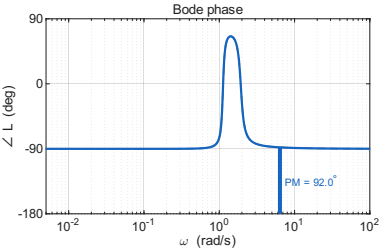
Case 22: two-inertia drive

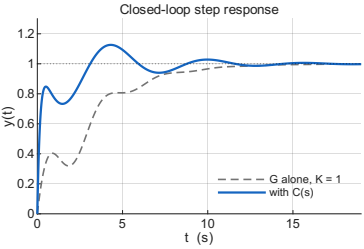
$K_v = 2 \to K = 6$ · $|L|$ crosses 0 dB 3 times · PM $= 92^\circ$ (highest crossing) · $M_p = 12.6\%$

Case 22: P design result









PI: a lag network with its pole at the origin

$$C(s) = K_p\left(1 + \frac{1}{T_i s}\right), \qquad \angle C(j\omega) = -\tan^{-1}\frac{1}{\omega T_i}$$

Procedure

(1) evaluate $G(j\omega_{gc})$ $\rightarrow$ (2) start with $\phi_{PI} = 5.71^\circ$, $T_i = 10/\omega_{gc}$ $\rightarrow$ (3) $K_p = 1/(|G|\,|1 + 1/(j\omega_{gc}T_i)|)$ $\rightarrow$ (4) GM too small $\rightarrow$ increase ϕ_{PI} (reduce T_i)

Case 01: $G = 1/(s+1)$

$\omega_{gc} = 2$: $|G| = 0.4472$, $\angle G = -63.4^\circ$ $\cdot$ $T_i = 5$, $K_p = 2.225$ $\rightarrow$ PM $= 110.9^\circ$
$t_s = 18.5$ s because the closed-loop pole at -0.1445 sits close to the PI zero

PI on a dead-time process: GM is the limit

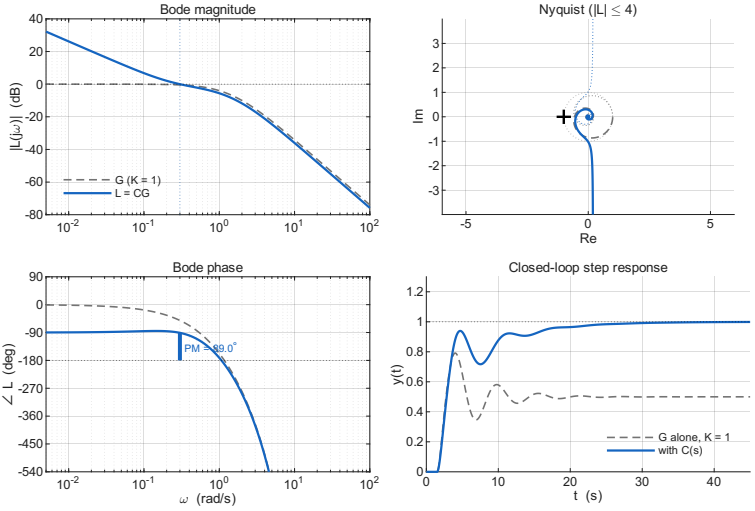
$$G = \frac{2e^{-1.5s}}{(s+1)(s+2)}, \qquad \omega_{gc} = 0.3$$

- First try $T_i = 33.3$: PM 123° but GM 4.51 dB $\times$
- Increase ϕ_{PI} to $40.03^\circ \to T_i = 3.968$, $K_p = 0.8084$
- PM $= 89.0^\circ$, GM $= 6.00$ dB $\checkmark$

Lesson

On delay-dominant processes the phase falls fast and the phase crossover sits near $\omega_{gc} \to$ check GM before PM

Case 16: PI on a dead-time process



Phase-lag compensation

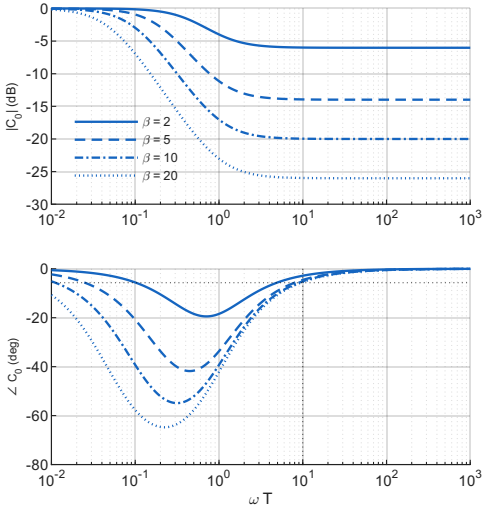
$$C(s) = K_c \frac{Ts + 1}{\beta Ts + 1}, \quad \beta > 1$$

It is the PI controller with its integrator moved off the origin. It works through attenuation $1/\beta$ at high frequency, not through its phase

Procedure

1. K_c from e_{ss}
2. Find ω_{gc} where $\angle G = -180^\circ + \text{PM} + 5.71^\circ$
3. $\beta = |K_c G(j\omega_{gc})|$
4. $T = 10/\omega_{gc}$

Frequency response of the lag network



Lag example: $G = 2/[(s+1)(s+2)]$, $e_{ss} \le 0.02$

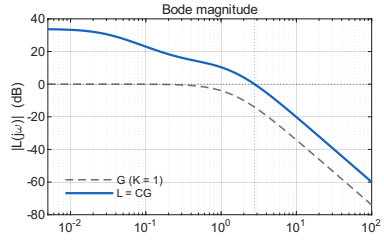
- $K_c = 49 \to \text{PM}_0$ is only 17.4°
- $\angle G = -124.3^\circ$ at $\omega_{gc} = 2.768 \to \beta = 9.75$, $T = 3.613$

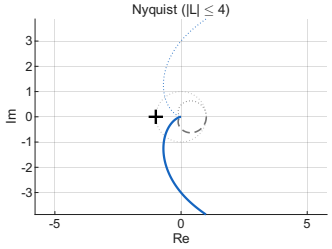
$$C = 49\,\frac{3.613s + 1}{35.22s + 1} \qquad \text{PM } 50.5^\circ \;\cdot\; M_p = 13\% \;\cdot\; t_s = 6.9\,\text{s}$$

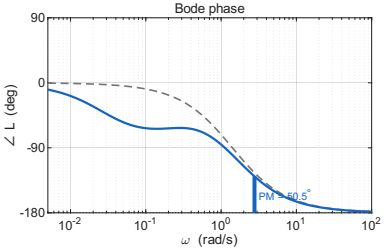
The price

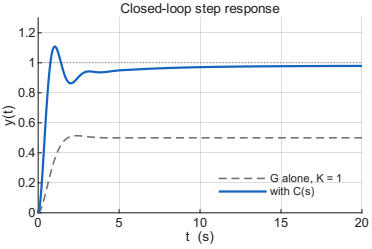
Bandwidth drops from 15.3 to 4.4 rad/s and a slow tail appears. P, PI and lag only work when $\phi_{req} \le 0$

Case 07: lag design result









Phase-lead compensation

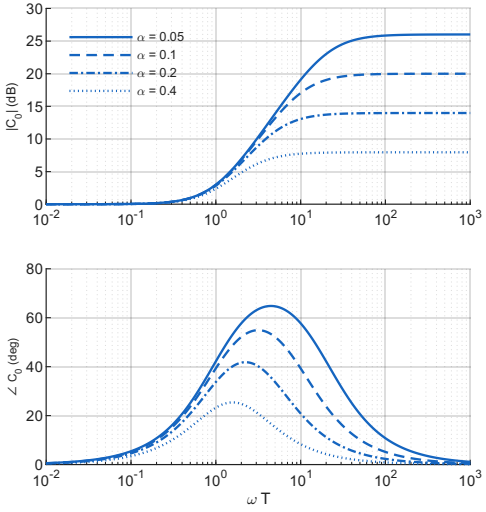
Lead is the mirror image of lag: lag buys accuracy with speed, lead buys phase — and speed — with high-frequency gain

$$C(s) = K_c \frac{Ts+1}{\alpha Ts+1}, \quad 0 < \alpha < 1$$

$$\sin\varphi_m = \frac{1-\alpha}{1+\alpha}, \qquad \omega_m = \frac{1}{T\sqrt{\alpha}}, \qquad |C_0(j\omega_m)| = \frac{1}{\sqrt{\alpha}}$$

- Phase is paid for with high-frequency gain $1/\alpha$
- One stage is practical up to $\varphi_m \approx 60$–65° ($\alpha \gtrsim 0.05$)

Frequency response of the lead network



Lead, Route A — accuracy first

1. K_c from e_{ss}; find PM_0 of K_cG
2. $\varphi_m = \text{PM} - \text{PM}_0 + \delta$ ($\delta \approx 5$–12° for the crossover shift)
3. Find α from $\sin\varphi_m = (1-\alpha)/(1+\alpha)$
4. New crossover where $|K_cG| = -10\log(1/\alpha)$ dB $\rightarrow \omega_m$
5. $T = 1/(\omega_m\sqrt{\alpha})$

Lead, Route B — speed first

1. $\varphi_m = \phi_{\text{req}}$; find α
2. Place $\omega_m = \omega_{\text{gc}}$, $T = 1/(\omega_{\text{gc}}\sqrt{\alpha})$
3. $K_c = \sqrt{\alpha}/|G(j\omega_{\text{gc}})|$

Limit of a single lead stage

$$\text{PM}_{\max}(\omega_c) = 180^\circ + \angle G + \sin^{-1}\frac{1-|K_cG|^2}{1+|K_cG|^2}$$

Example: position servo $G = 1/[s(s+1)]$, $K_v = 10$

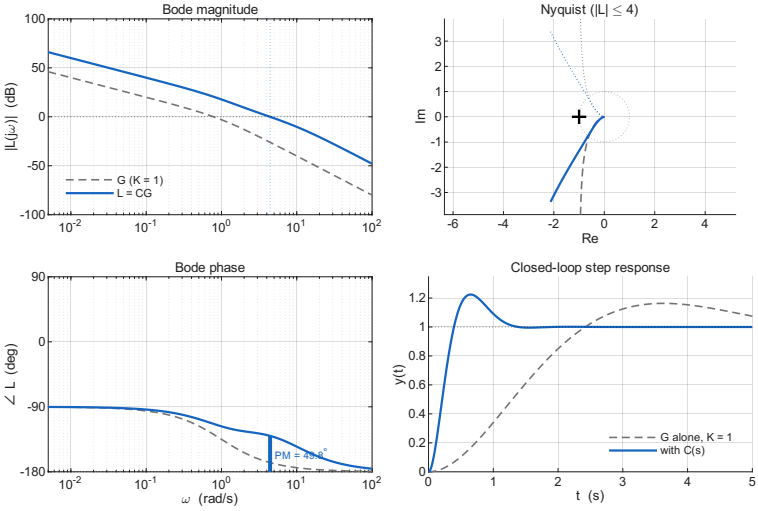
1. $K_c = 10 \to \mathsf{PM}_0 = 17.96^\circ$
2. $\varphi_m = 37.04^\circ$, $\alpha = 0.2482$
3. $-6.05\,\mathsf{dB} \to \omega_m = 4.425$
4. $T = 0.4536$

$$C = 10\,\frac{0.4536s + 1}{0.1126s + 1}$$

Result

PM $= 49.8^\circ$ · $M_p = 22.4\%$ · $t_s = 1.2\,\mathsf{s}$ · crossover more than 5 times faster

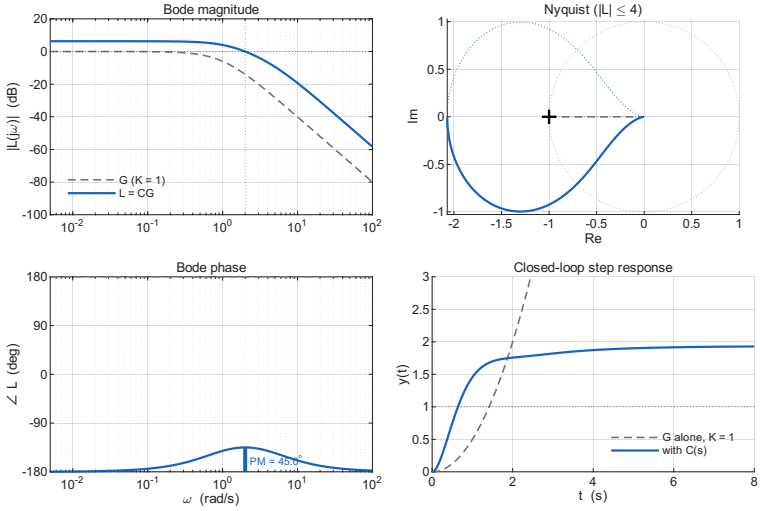
Case 11: lead design result



Lead with an RHP pole: magnetic levitator $G = 1/(s^2 - 1)$

- Phase is -180° at every frequency $\rightarrow$ gain alone cannot help $\cdot$ need $\omega_{gc} \gtrsim 2p = 2$
- $\varphi_m = 45^\circ \rightarrow C = 2.071 \dfrac{1.207s + 1}{0.2071s + 1}$
- Nyquist encircles -1 once counter-clockwise $\rightarrow Z = 0$
- Lower GM -6.32 dB: reducing the gain by more than this destabilises the loop
- $y(\infty) = 1.934 \rightarrow$ a pre-filter is required

Case 15: lead on an unstable plant



Lag–lead: accurate *and* fast

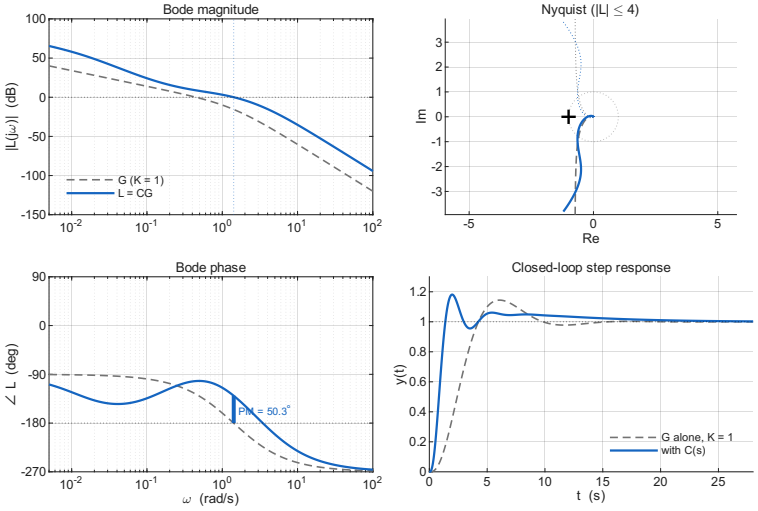
$G = 1/[s(s+1)(s+2)]$, $K_v = 10 \to K_c = 20$ (closed loop unstable)

1. $\omega_{\text{gc}} = \omega_{pc}(G) = \sqrt{2}$
2. Lead: $\varphi_m = 55.71^\circ$, $\alpha = 0.0952$
3. Lag: $\beta = |K_c G|/\sqrt{\alpha} = 10.81$
4. $T_2 = 10/\omega_{\text{gc}}$

Result

PM 50.3° · GM $12.4\,\text{dB}$ · $M_p = 18\%$

Case 18: lag–lead design result



PD and PID: a lead network with $\alpha = 1/N$

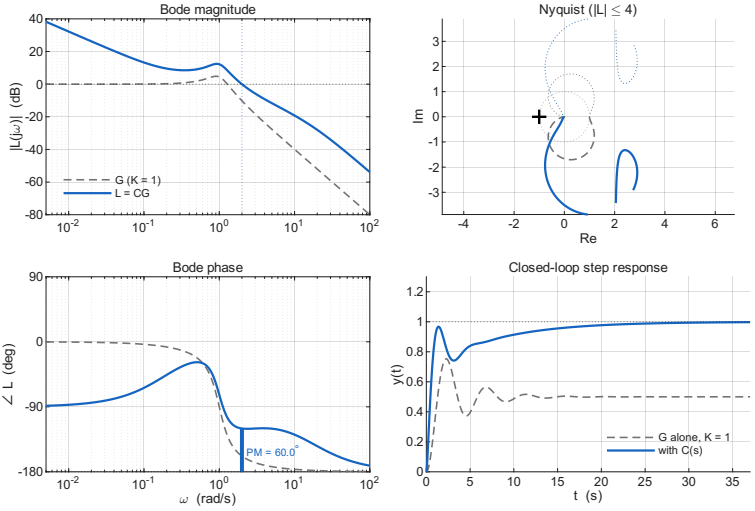
$$C = K_p\Big(1 + \frac{1}{T_i s}\Big)\frac{T_d s + 1}{(T_d/N)s + 1}, \qquad \phi_{D,\max} = \sin^{-1}\frac{N-1}{N+1} = 54.9^\circ \ (N = 10)$$

1. $T_i = 10/\omega_{gc}$
2. $\phi_D = -180^\circ + \text{PM} - \angle G + 5.71^\circ$
3. Solve T_d from ϕ_D, then K_p

Case 09: $G = 1/(s^2 + 0.6s + 1)$

$\omega_{gc} = 2$: $\angle G = -158.2^\circ \to \phi_D = 43.9^\circ$ · $T_d = 0.616$, $K_p = 2.042 \to$ PM 60°
$|u|_{\max} = 20.4$ from derivative kick $\to$ use 2-DOF (Chapter 12)

Case 09: PID design result



Analytical design at a single frequency

$$C(j\omega_c) = \frac{e^{j\theta}}{|G(j\omega_c)|}, \qquad \theta = -180^\circ + \text{PM} - \angle G(j\omega_c)$$

First-order compensator $C = \dfrac{a_1 s + a_0}{b_1 s + 1}$

$$a_1 = \frac{1 - a_0|G|\cos\theta}{\omega_c |G| \sin\theta}, \qquad b_1 = \frac{\cos\theta - a_0|G|}{\omega_c \sin\theta}, \qquad 0 < a_0 < \frac{\cos\theta}{|G|}$$

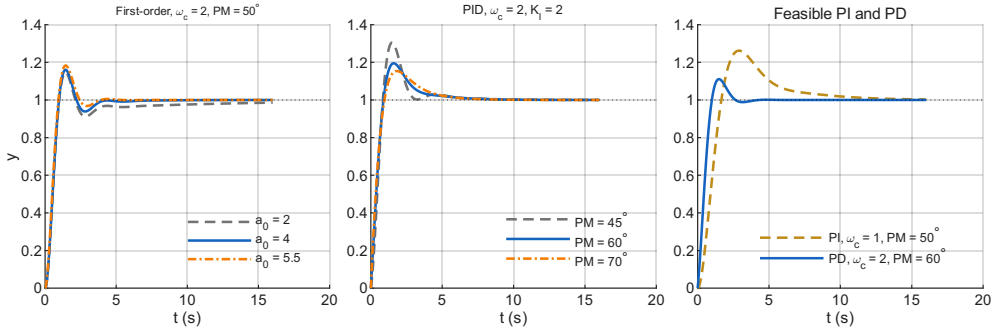
$G = 1/[s(s+2)]$, $\omega_c = 2$, PM 50°, $a_0 = 4 \to C = \dfrac{9.5925s + 4}{1.6585s + 1}$ (PM $= 50.0^\circ$ exactly)

Analytical design: ideal PID

$$K_P = \frac{\cos\theta}{|G|}, \qquad K_D = \frac{\sin\theta}{\omega_c |G|} + \frac{K_I}{\omega_c^2}$$

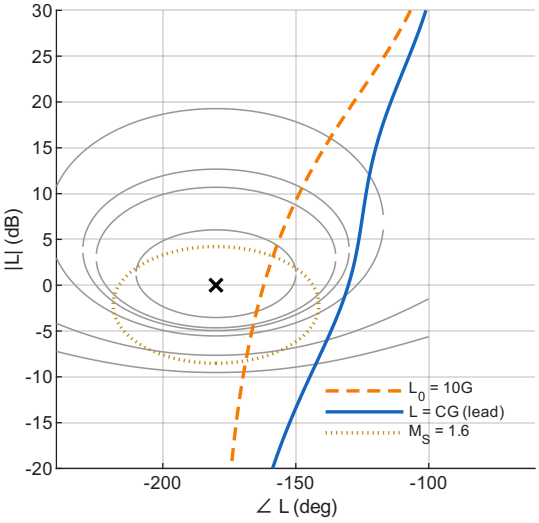
- K_I is a free parameter (set it from the accuracy requirement)
- PI is feasible when $\theta < 0$ · PD is feasible when $\theta > 0$
- The formula only guarantees the point ω_c — always verify stability and GM

The third parameter still matters



- $a_0 = 2$ vs 5.5: same PM, but t_s differs by a factor of 3.6
- Higher PM $\rightarrow$ lower M_p, but t_s does not drop (slow real pole)
- PI at $\omega_c = 2$, PM $60°$: $K_I < 0 \rightarrow$ infeasible

The Nichols chart



Reading the Nichols chart

- Axes: $|L|$ (dB) versus $\angle L$ (deg) $\cdot$ contours of constant $|T| = M$
- Changing the gain shifts the locus vertically
- Design: make the locus touch the target M_T contour and stay outside the M_S contour

Case 11

Before: $M_T = 10.11$ dB, $M_S = 3.35$ $\quad\rightarrow\quad$ after lead: $M_T = 1.95$ dB, $M_S = 1.42$

Part 3

Fundamental Limitations

Bode's gain–phase relation

$$\angle G(j\omega_0) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{dM}{du}\,\ln\coth\frac{|u|}{2}\,du$$

(minimum-phase plant) slope $-20n$ dB/dec $\rightarrow$ phase $\approx -90n^\circ$

Design consequence

Choosing the slope at crossover = choosing the PM · aim for ≈ -20 dB/dec over about a decade around ω_{gc}

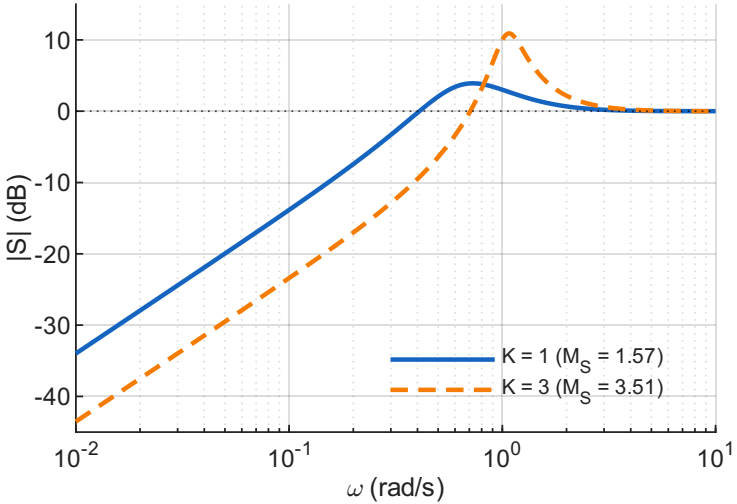
- 50% of the kernel weight lies within ± 0.23 decade
- Lead does not "create phase"; it *flattens the slope*
- RHP zeros and dead time are all-pass: unit magnitude but negative phase $\rightarrow$ more damaging

Bode's sensitivity integral: the waterbed effect

$$\int_0^\infty \ln|S(j\omega)|\,d\omega = \pi \sum \mathrm{Re}\{p_k\}$$

- Push $|S|$ down in one band and it bulges up elsewhere
- $L = K/[s(s+1)(s+2)]$: the integral is $< 10^{-12}$ for every K
- $K : 1 \to 3 \to M_S : 1.57 \to 3.51$
- Unstable plant: the integral equals $\pi \sum \mathrm{Re}\{p_k\} > 0$ — a debt that can never be repaid

The waterbed effect: $|S|$ as K increases



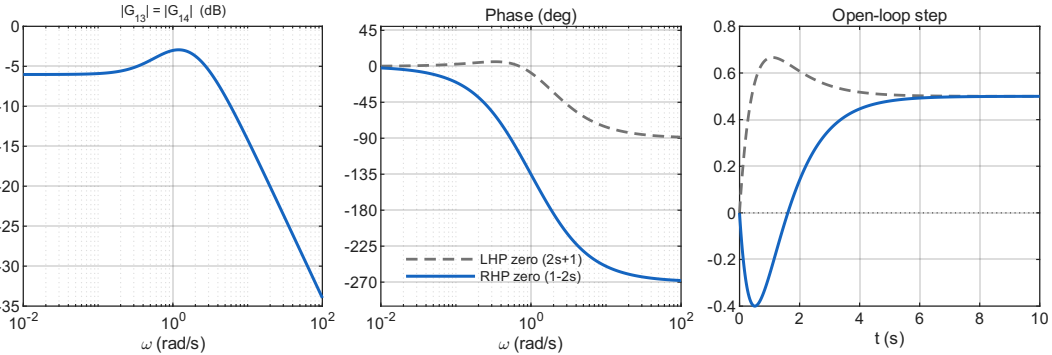
Ceilings from RHP zeros and dead time

$$\omega_{gc} \le z \tan\Big(\frac{90^\circ - \text{PM}}{2}\Big), \qquad \omega_{gc}\theta \le \frac{\pi}{2} - \text{PM (rad)}, \qquad \omega_{gc} \gtrsim 2p$$

PM	RHP zero	delay $\omega_{gc}\theta$
30°	$0.577z$	1.047
45°	$0.414z$	0.785
60°	$0.268z$	0.524

A plant with both p and z needs $z \gtrsim 4p$ for a usable range of ω_{gc}

RHP zero versus a minimum-phase plant



Same magnitude at every frequency, but the RHP zero loses phase and responds in the wrong direction first

The universal FOPDT table: $L = x\,e^{-s}/s$, $x = \omega_{gc}\theta$

x	PM	GM dB	M_S	$M_p\%$	t_s/θ
0.30	72.8	14.38	1.307	0.0	9.35
0.40	67.1	11.88	1.438	0.1	5.43
0.50	61.4	9.94	1.591	4.1	6.07
0.60	55.6	8.36	1.773	11.8	5.69
0.785	45.0	6.02	2.232	28.9	9.01
1.00	32.7	3.92	3.129	50.3	12.94

- PI with $T_i = \tau$ on an FOPDT plant gives exactly this loop · every property depends on x alone
- Minimum $t_s/\theta \approx 4.0$ at $x \approx 0.46$ (slight overshoot) · $M_S = 1.6$ at $x = 0.506$

How fast can the TCLab heater be?

$$G = \frac{1.2\,e^{-30s}}{150s + 1} \qquad \text{specification } M_S \le 1.6$$

(a) 1-DOF PI

From the table $x = 0.4625 \to t_s/\theta = 4.05 \to t_s \approx 121$ s

(b) Any controller

Best case $T = e^{-\theta s}/(\lambda s + 1)$; $\lambda \to 0$ gives $M_S \to 2$ · $M_S = 1.6 \to \lambda = 19.91$ s $\to$ $t_s \ge 30 + 19.91 \ln 50 = 107.9$ s

Lesson

No structure (PID, MPC, neural network) settles faster than 108 s with $M_S \le 1.6$ — the 30 s delay sets the limit, not the designer

IMC (λ) tuning

$G = G_- G_+,\ G_- = K/(\tau s + 1),\ G_+ = e^{-\theta s}$ · target $T = G_+/(\lambda s + 1)$ with $e^{-\theta s} \approx 1 - \theta s$

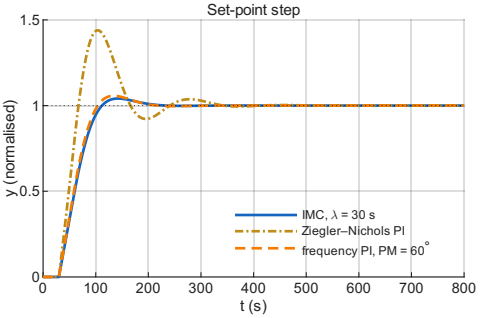
IMC PI

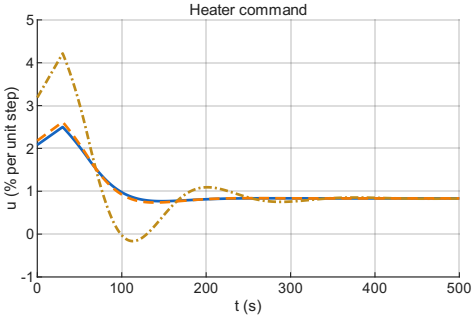
$$K_c = \frac{\tau}{K(\lambda + \theta)}, \qquad T_i = \tau$$

- $\omega_{gc} = 1/(\lambda + \theta) \to x = \theta/(\lambda + \theta)$
- Rule $\lambda = \theta \to x = 0.5$, PM 61.4°, $M_S = 1.59$ — the delay ceiling in disguise
- One knob: increasing λ always increases robustness

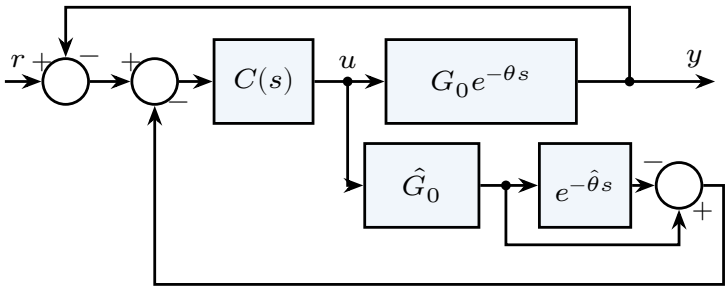
IMC vs Ziegler–Nichols vs frequency-domain PI

Tuning	K_c	T_i	PM	GM dB	M_S	$M_p\%$	t_s	load dev.
IMC $\lambda = 30$	2.083	150	61.4	9.94	1.591	4.1	182	3.51
Z–N PI	3.188	93.0	36.2	5.67	2.427	43.8	312	3.07
Frequency PM 60°	2.182	150	60.0	9.54	1.631	5.7	181	3.46





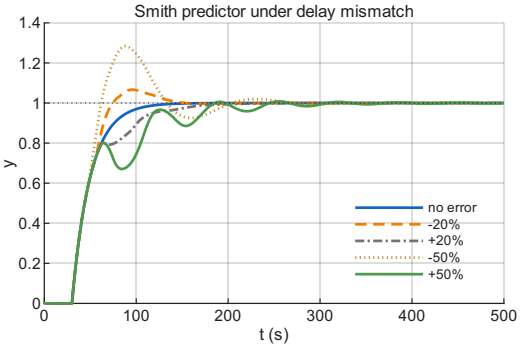
The Smith predictor

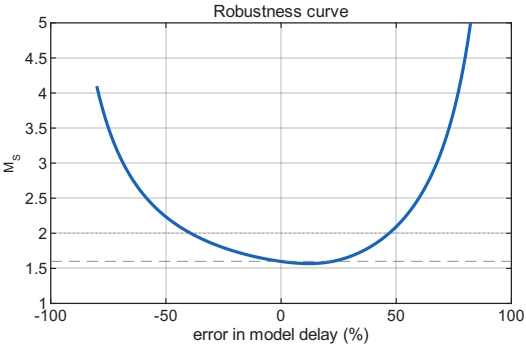


$$T = \frac{CG_0}{1 + CG_0}\, e^{-\theta s} \quad (\hat{G}_0 = G_0,\ \hat{\theta} = \theta)$$

- Design C on the delay-free plant; the result is simply delayed by θ
- TCLab: $\lambda = 20\,\text{s} \to K_c = 6.25$, $t_s = 108\,\text{s}$ reaches the limit · the same K_c without the predictor: $M_S = 26.2$!

The Smith predictor with delay mismatch





- $M_S > 1.6$ as soon as the model delay is shorter than the true delay · worst case at $\pm 50\%$: $M_S = 2.24$
- IMC PI with $\lambda = 45\,\text{s}$: $t_s = 163\,\text{s}$, worst case at $\pm 50\%$: $M_S = 1.77$ — better, and no model needed

Part 4

Choosing and Checking a Design

Selection rules from the Design Atlas of 25 plants

1. **Look at the phase at crossover first**: $\phi_{\text{req}} \leq 0 \rightarrow$ P/PI/lag $\cdot$ $\phi_{\text{req}} > 0 \rightarrow$ lead/PD/PID
2. **Accuracy conflicts with margins**: lag or lag–lead
3. **Type 2 / undamped**: unstable for every gain $\rightarrow$ phase lead is mandatory
4. **RHP zero / dead time**: PI at low ω_{gc}; GM is usually the binding constraint
5. **RHP pole**: ω_{gc} must exceed the pole, GM is a lower bound, a pre-filter may be needed
6. **Washout** ($G(0) = 0$): cannot track a step at all
7. **Multiple crossings**: check them all and confirm with Nyquist

Observation

Designs tuned to GM ≈ 6 dB on delay plants always land at $M_S \approx 2.0$–2.1 — because GM 6 dB and $M_S = 2$ are almost the same statement

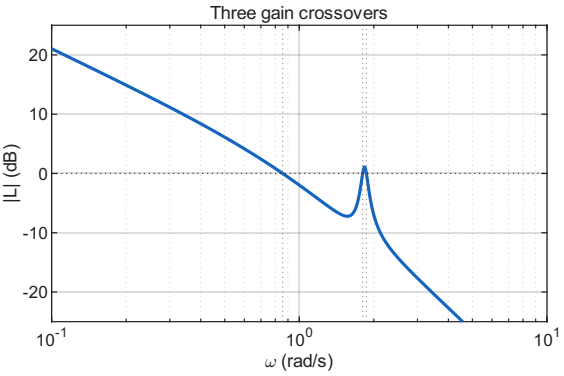
Counterexample: good margins, poor robustness

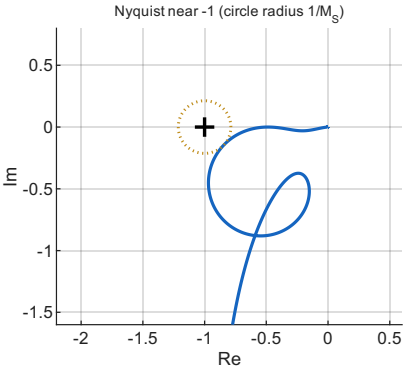
$$L = \frac{1.186(s^2 + 0.4078s + 3.2141)}{s(s+1)(s^2 + 0.1135s + 3.3648)}$$

- Reported: GM 6.14 dB, PM 55.3° ✓
- Reality: $M_S = 4.695$, three crossings · PM at the third crossing $= 18.6^\circ$
- Disk margin only ± 2.0 dB / $\pm 13.1^\circ$

Decide with M_S · report GM/PM alongside for continuity

Counterexample: Bode and Nyquist





Design checklist

	Check	Because
1	Is the specification achievable (delay/RHP-zero ceiling)?	No controller beats the ceiling
2	Report M_S, M_T, not just GM/PM	Margins probe only two directions
3	Check *every* gain crossover	Each has its own PM
4	DM against sampling/unmodelled delay	Sampling adds half a period of delay
5	Simulate the load response, not only set point	Loops fail through GS
6	Check $\lvert u\rvert_{\max}$ and noise into the actuator	An actuator that cannot follow is not a design
7	Robustness over the parameter range	The nominal case is the easiest
8	Simulate with saturation and anti-windup	Linear analysis ends at the actuator limits

Verifying in MATLAB

```
s = tf('s'); G = 1/(s*(s+2));
wc = 2; PM = 60; Ki = 2;
g = evalfr(G,1i*wc); M = abs(g); psi = angle(g);
th = -pi + PM*pi/180 - psi;
Kp = cos(th)/M;
Kd = sin(th)/(wc*M) + Ki/wc^2;
C = Kp + Ki/s + Kd*s; L = C*G;
assert(isstable(feedback(L,1))) % never implied
S = allmargin(L) % every crossover
Ms = norm(feedback(1,L),inf)
```

- `allmargin` reports *every* crossing (`margin` reports one) · `norm(S,inf)` $= M_S$
- Set `G.InputDelay` $\rightarrow$ exact delay, no Padé needed · all scripts are in the `matlab/` folder

Chapter summary

- Frequency-domain design = loop shaping of $|L(j\omega)|$ in three bands
- Specifications translate: $M_p \to \zeta \to$ PM, $t_s \to \omega_{gc}$, $e_{ss} \to K_c$ · judge robustness by M_S
- Group $\phi_{req} \le 0$: P shifts the magnitude · PI = lag with its pole at the origin · lag trades speed for accuracy
- Group $\phi_{req} > 0$: lead buys phase with high-frequency gain · lag–lead combines both · PD = lead with $\alpha = 1/N$
- Analytical formula $C(j\omega_c) = e^{j\theta}/|G|$ fixes one frequency — always verify
- Bode: choosing the slope = choosing the PM · waterbed: sensitivity can be moved, not removed
- Ceilings: $\omega_{gc} \le z\tan[(90^\circ - \text{PM})/2]$, $\omega_{gc}\theta \le \pi/2 - \text{PM}$, $\omega_{gc} \gtrsim 2p$
- FOPDT: IMC $K_c = \tau/[K(\lambda+\theta)]$, $T_i = \tau$ · the Smith predictor is sensitive to delay mismatch

Assignments

Textbook exercises

Theory: 11.1–11.5, 11.22–11.25 · case-study design: 11.6–11.20
Process tuning: 11.26–11.27 · analytical: 11.30–11.31

Worksheet 11

Design P, PI, lag, lead×2, lag–lead, PD and PID for $20/[s(s+1)(s+2)]$ and compare ω_{gc}, PM, GM, M_S, t_s

TCLab laboratory

Exercise 11.21: identify a model → tune PI → test on the hardware